

AQI Predictor Project

Milestone 1 Report: Project Setup & API Exploration

Ayesha Salahuddin

GitHub: github.com/ayeshasalahuddin/AQI-predictor

Abstract—This report documents the complete environment setup, tool configuration, and API verification carried out in Milestone 1 of the Pearls AQI Predictor project. The goal of this milestone was to establish a stable, reproducible development foundation before any machine learning work begins. All required tools were installed, cloud accounts were created and connected, and a live AQI reading of 161 was successfully fetched from the AQICN API for Karachi, Pakistan, confirming the data access layer is fully operational.

Index Terms—AQI, air quality, environment setup, Hopsworks, AQICN, Python, conda, feature store, version control

I. INTRODUCTION

The Pearls AQI Predictor is an end-to-end machine learning system that predicts the Air Quality Index (AQI) of Karachi for the next 3 days using a 100% serverless architecture. The system is built on four core pillars:

- **Feature Pipeline:** Fetches live weather and pollution data every hour, engineers features, and stores them in a cloud Feature Store
- **Training Pipeline:** Trains and evaluates multiple ML models daily and registers the best one
- **CI/CD Automation:** GitHub Actions runs both pipelines automatically on schedule
- **Web App Dashboard:** A Streamlit application displaying real-time and forecasted AQI with health alerts and model explainability

The project uses **Hopsworks** as the central Feature Store and Model Registry, **AQICN** as the live data source, and **GitHub** for version control and automation.

Milestone 1 focused on establishing the complete foundation for the project. A stable, reproducible foundation is critical — without it, every subsequent milestone would be fragile and error-prone. This report documents every step taken, the reasoning behind each decision, and the verified output confirming the foundation is solid.

II. MILESTONE OBJECTIVE

The objective of Milestone 1 was to set up every tool, account, and environment needed before any machine learning work begins. By the end of this milestone, the following was verified to be working:

- A live AQI reading of **161** was successfully fetched from the AQICN API for Karachi
- All cloud accounts (GitHub and Hopsworks) were connected and operational

- The development environment was reproducible and isolated via conda
- All project files were version-controlled and pushed to GitHub

III. TOOLS AND TECHNOLOGIES USED

Tool	Purpose	Status
Python 3.11	Core programming language	Verified
VS Code	Code editor	Installed
Git & GitHub	Version control & hosting	Repo pushed
Conda	Environment management	aqi env active
requests	HTTP calls to APIs	Installed
pandas 2.3.3	Data manipulation	Installed
python-dotenv	Secure key management	Installed
hopsworks 4.7.4	Feature Store & Model Registry	Account created
AQICN API	Live AQI data source	Token tested

IV. IMPLEMENTATION

A. Development Environment

Python 3.11 was installed and a conda virtual environment named `aqi` was created to isolate all project dependencies. A conda environment was chosen over the standard Python `venv` tool because the machine lacked the C++ build tools (Ninja/CMake) required to compile pandas and numpy from source. Anaconda ships pre-compiled binary wheels for Windows, eliminating this dependency entirely.

B. Version Control Setup

A public GitHub repository named `AQI-predictor` was created under the account `ayeshasalahuddin`. The repository was initialized with a `README`, a Python `.gitignore`, and an MIT license. The repo was cloned locally to the Desktop and a `milestone1` branch was created to isolate all Milestone 1 work.

C. Credential Management

All API keys are stored in a local `.env` file that is excluded from version control via `.gitignore`. The `python-dotenv` library loads them at runtime:

```
load_dotenv()
TOKEN = os.getenv("AQICN_TOKEN")
```

This ensures credentials are never exposed in the codebase or on GitHub.

D. Cloud Account Setup

A free Hopsworks account was created at `app.hopsworks.ai` and a project named `PredictorAQI` was initialized. A Hopsworks API key was generated and stored in the `.env` file. Hopsworks will serve as the central Feature Store and Model Registry for all subsequent milestones.

E. API Verification

A test script `test_api.py` was written to verify end-to-end connectivity with the AQICN API:

```
url = f"https://api.waqi.info/feed/{CITY}/?token={TOKEN}"
response = requests.get(url)
data = response.json()

if data["status"] == "ok":
    aqi = data["data"]["aqi"]
    print(f"Current_AQI:_{aqi}")
```

V. STEPS COMPLETED

#	What Was Done	Why
1	Installed Python 3.11	Core language
2	Installed VS Code	Code editor
3	Created GitHub repo	Version control
4	Cloned repo to Desktop	Local development
5	Created conda env 'aqi'	Isolated dependencies
6	Installed all libraries	Pipeline requirements
7	Created .env file	Secret management
8	Added .env to .gitignore	Credential security
9	Created Hopsworks account	Feature Store access
10	Obtained AQICN token	Data source access
11	Wrote and ran test_api.py	End-to-end verification
12	Pushed work to GitHub	Cloud backup

VI. TECHNICAL CHALLENGES AND RESOLUTIONS

Issue	Cause	Resolution
venv activation failed	PowerShell execution policy	Used <code>.\venv\bin\activate</code>
pandas build failure	Missing Ninja/CMake on Windows	Switched to Anaconda
conda not on PATH	Init not run	Ran <code>conda init powershell</code>

VII. RESULTS AND VERIFICATION

The following output was produced by running `test_api.py`, confirming the entire data access layer is functional:

```
City: Karachi US Consulate, Pakistan
Current AQI: 161
Raw data keys: ['aqi', 'idx', 'attributions', 'city', 'dominantpol', 'iaqi', 'time', 'forecast', 'debug']
```

An AQI of **161** falls in the **Unhealthy** category on the US EPA scale (151–200), meaning air quality in Karachi at the time of testing was poor enough to affect all individuals. This confirms the API is live, the token is valid, and the data contains all fields needed for the feature pipeline in Milestone 2.

VIII. NEXT STEPS — MILESTONE 2

Milestone 2 will build the Feature Pipeline script (`feature_pipeline.py`). This script will:

- Fetch raw weather and pollutant data from the AQICN API
- Engineer time-based features (hour, day of week, month) and derived features (AQI change rate, rolling averages)
- Define targets — AQI values 24, 48, and 72 hours into the future
- Connect to Hopsworks and write all features as versioned rows into the Feature Store

Upon completion of Milestone 2, the project will have a live running data pipeline that continuously populates the Feature Store with structured, ML-ready data.

IX. CONCLUSION

Milestone 1 successfully established the complete development foundation for the AQI Predictor project. All tools were installed, cloud accounts were connected, credentials were secured, and live API data was verified. The project is version-controlled on GitHub under a dedicated `milestone1` branch and is fully ready for the feature engineering work of Milestone 2.